C. Matching Arrays

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: binary search, constructive algorithms, greedy, sortings | https://codeforces.com/problemset/problem/1896/C

You are given two arrays a and b of size n. The *beauty* of the arrays a and b is the number of indices i such that a_i gt b_i.

You are also given an integer x. Determine whether it is possible to rearrange the elements of b such that the beauty of the arrays becomes x. If it is possible, output one valid rearrangement of b.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains two integers n and x ($1 \le n \le 2 \cdot 10^5$, $0 \le x \le n$) — the size of arrays a and b and the desired beauty of the arrays.

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 2n$) — the elements of array a.

The third line of each test case contains n integers $b_1, b_2, \ldots, b_n$ ($1 \le b_i \le 2n$) — the elements of array b.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output "NO" if it is not possible to rearrange b to make the beauty of the arrays equal to x.

Otherwise, output "YES". Then, on the next line, output n integers which represent the rearrangement of b.

If there are multiple solutions, you may output any of them.

You can output "YES" and "NO" in any case (upper or lower). For example, the strings "yEs", "yes", "Yes", and "YES" will be recognized as positive responses.

Example

Input

```
7
1 0
1
2
1 1
1
2
3 0
2 4 3
4 1 2
3 1
2 4 3
4 1 2
3 2
2 4 3
4 1 2
3 3
2 4 3
4 1 2
5 2
6 4 5 6 2
9 7 9 1 1
```

Output

```
YES
2
NO
NO
YES
2 4 1
YES
4 1 2
NO
YES
1 9 9 7 1
```

Note

In test cases 1 and 2, the beauty of the arrays has to be 0 since $a_1 = 1 \leq 2 = b_1$.

In test cases 3, 4, 5 and 6, the only possible beauty of the arrays is $x = 1$ and $x = 2$. In particular, if b is rearranged to [2, 4, 1], then $a_3 = 3$ gt $1 = b_3$, so the beauty of the arrays is 1. If b is kept in the same order as given the input, then $a_2 = 4$ gt $b_2 = 1$ and $a_3 = 3$ gt $2 = b_3$, so the beauty of the arrays is 2.